

1. Logic, Sets and Numbers

1.1 Logic

Def. 1. A **mathematical Statement** is a well-formulated assertion that is strictly either true or false.

Def. 2. A **Statement** that depends on one or multiple variables is called a **predicate** and denoted by $P(x), Q(x), \dots$.

Def. 3. **Quantifiers** for predicates:

- $\forall x \in \mathbb{R} : P(x)$ (for all x holds $P(x)$)
- $\exists x \in \mathbb{R} : P(x)$ (there exists x s.t. $P(x)$)
- $\exists! x \in \mathbb{R} : P(x)$ (exists exactly one x)
- $\nexists x \in \mathbb{R} : P(x)$ (exists no x s.t. $P(x)$)

Imp. 4. The order of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.

Def. 5. Basic **logical connectives** are defined as:

$$\begin{aligned} A \wedge B &:= A \text{ and } B \text{ are true} \\ A \vee B &:= A \text{ or } B \text{ is true} \\ A \implies B &:= A \text{ implies } B \text{ is true} \\ A \iff B &:= A \text{ is equivalent to } B \text{ is true} \end{aligned}$$

Lem. 6.

$$\neg A \vee B \iff A \implies B$$

Thm. 7. The following equivalences hold:

$$\begin{aligned} \neg(\forall x P(x)) &\iff \exists x \neg P(x) \\ \neg(\exists x P(x)) &\iff \forall x \neg P(x) \end{aligned}$$

as well as

$$\begin{aligned} \forall x (P(x) \wedge Q(x)) &\iff (\forall x P(x)) \wedge (\forall x Q(x)) \\ \exists x (P(x) \vee Q(x)) &\iff (\exists x P(x)) \vee (\exists x Q(x)) \end{aligned}$$

1.2 Sets

Def. 8. A **set** is a collection of elements, that are listed or characterized by a certain property. The empty set $:= \emptyset$ and is the only set that contains no elements.

Def. 9. Let A and B be sets. A is a **subset** of B denoted by $A \subset B$ iff.

$$\begin{aligned} \forall x (x \in A \implies x \in B) \\ A \text{ is a **proper subset** of } B \text{ denoted by } A \subsetneq B \text{ iff.} \\ A \subset B \wedge A \neq B \end{aligned}$$

Def. 10. Let A and B be sets. The **union** of A and $B := A \cup B := \{x \mid x \in A \vee x \in B\}$.

Def. 11. The **intersection** of A and $B := A \cap B := \{x \mid x \in A \wedge x \in B\}$.

Def. 12. The **difference** of A and $B := A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.

Def. 13. The **Cartesian product** of A and $B := A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.

Rem. 14. Note that if $B \subset A$ then the difference $A \setminus B$ is called the complement of B in A and $:= B^c$.

1.3 Numbers

Rem. 15. Some important subsets of the real numbers are

$$\begin{aligned} \mathbb{R}_0^+ / \mathbb{R}^+ \\ \mathbb{R}_0^- / \mathbb{R}^- \end{aligned}$$

which define the intervals

$$\begin{aligned} [0, \infty) / (0, \infty) \\ (-\infty, 0] / (-\infty, 0) \end{aligned}$$

Def. 16. Continuous subsets are called **intervals**.

Def. 17. An interval is called **open** if it does not contain its endpoints

$$\begin{aligned} I = (a, b) &:= \{x \mid a < x < b\} \\ \text{closed if it contains its endpoints} \\ I = [a, b] &:= \{x \mid a \leq x \leq b\} \\ \text{half-open if it contains one endpoint and not the other} \\ I = [a, b) &:= \{x \mid a \leq x < b\} \end{aligned}$$

Rem. 18. An arbitrary subset of $\mathbb{R}$ is called open if it is a union of open intervals and closed if its complement is open.

Def. 19. An **upper (lower) bound** of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:

$$\begin{aligned} \forall x \in X : x \leq (\geq) y \\ \text{The **maximum (minimum)** of a subset } X \subset \mathbb{R} \text{ is the element } x_0 \in X \text{ such that:} \\ \forall x \in X : x \leq (\geq) x_0 \\ \text{Often denoted as **max(X)** and **min(X)**.} \end{aligned}$$

Def. 20. An interval $I = (a, b)$ is **bounded** if there exist upper and lower bounds hence

$$a > -\infty \quad \text{and} \quad b < \infty$$

Rem. 21. A bounded and closed interval is often referred to as **compact**.

Def. 22. The **Supremum** $\sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of X . and can be characterized by:

$$\begin{aligned} \forall x \in X : x \leq \sup(X) \wedge \\ \forall \epsilon > 0 : \exists x \in X : x > \sup(X) - \epsilon \end{aligned}$$

Def. 23. The **Infimum** $\inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of X . and can be characterized by:

$$\begin{aligned} \forall x \in X : x \geq \inf(X) \wedge \\ \forall \epsilon > 0 : \exists x \in X : x < \inf(X) + \epsilon \end{aligned}$$

Thm. 24. The **maximum and minimum are unique** as long as they exist.

Def. 25. **Order Axioms** reflexivity:

$$\forall x \in X : x \leq x$$

antisymmetry:

$$\forall x, y \in X : x \leq y \wedge y \leq x \implies x = y$$

transitivity:

$$\forall x, y, z \in X : x \leq y \wedge y \leq z \implies x \leq z$$

totality:

$$\forall x, y \in X : x \leq y \vee y \leq x$$

Def. 26. A set X is said to be **(order)complete** iff for every non empty subsets $A, B \subseteq X$

$$\begin{aligned} \forall a \in A \forall b \in B : a \leq b \\ \implies \exists c \in X \quad \text{s.t.} \\ \forall a \in A : a \leq c \wedge \\ \forall b \in B : c \leq b \end{aligned}$$

Ex. 27. The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition, multiplication, distributivity and order. Furthermore $\mathbb{R}$ is an order complete field.

Ex. 28. $\mathbb{Q}$ is not order complete. Let $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ and $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$. Then $\forall a \in A \forall b \in B : a \leq b$ but there is no $c \in \mathbb{Q}$ such that $c^2 = 2$.

Thm. 29. Every **non empty, bounded from above (below) subset** $X \subset \mathbb{R}$ has a **supremum (infimum)**.

Thm. 30. The **Archimedian Principle** says that For $x \in \mathbb{R}$ and $y > 0$ there exists an element

$$\begin{aligned} n \in \mathbb{N} \quad \text{s.t.} \quad n \cdot y > x \\ \text{For every } \epsilon > 0 \text{ there exists an element } n \in \mathbb{N} \text{ such that} \\ \frac{1}{n} < \epsilon \end{aligned}$$

Thm. 31. The Young Inequality states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

1.3.1 Vectors and Vector Spaces

Def. 32. The **euclidian space** $\mathbb{R}^n$ is the set of all vectors $x = (x_1, x_2, \dots, x_n)$ where $x_i \in \mathbb{R}$.

Def. 33. **Addition and scalar multiplication** are defined as:

$$\begin{aligned} x + y \\ = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\ \lambda x \\ = (\lambda x_1, \lambda x_2, \dots, \lambda x_n) \end{aligned}$$

Def. 34. A **vector space** must satisfy the following axioms:

Commutativity: $x + y = y + x$

Associativity: $x + (y + z) = (x + y) + z$

Identity: $\exists e \in X$ s.t. $x + e = x \quad \forall x \in X$

Inverse: $\forall x \in X \exists x^{-1} \in X$ s.t. $x + x^{-1} = e$

Distributivity: $x \cdot (y + z) = x \cdot y + x \cdot z$

Def. 35. For $x, y \in \mathbb{R}^n$, the **standard scalar product** is $\langle x, y \rangle :=$

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^n x_i y_i$$

The **euclidean norm** is:

$$\|x\| := \sqrt{\sum_{i=1}^n x_i^2}$$

The **euclidean distance** between x and y is:

$$d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Thm. 36. The triangle- and the cauchy-schwarz inequality hold for all $x, y, z \in \mathbb{R}^n$:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

1.3.2 Complex Numbers

Def. 37. The **complex numbers** $\mathbb{C}$ are the set of numbers

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

where a is the **real part** $\operatorname{Re}(z)$, and b is the **imaginary part** $\operatorname{Im}(z)$.

Def. 38. Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$: Addition:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

Multiplication:

$$z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$$

Def. 39. The **complex conjugate** of $z = x + iy$ is $\bar{z} = x - iy$.

Imp. 40. Division is calculated by expanding with the conjugate of the denominator:

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Lem. 41. Some useful properties of the conjugate:

$$z \cdot \bar{z} = |z|^2$$

$$z + \bar{z} = 2\operatorname{Re}(z)$$

$$z - \bar{z} = 2i\operatorname{Im}(z)$$

$$\overline{z + w} = \bar{z} + \bar{w}$$

$$\overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

Def. 42. Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of $z = a + ib$ is its **distance from the origin**:

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$.

Thm. 43. The Euler's formula states that for $t \in \mathbb{R}$:

$$e^{it} = \cos(t) + i \sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Rem. 44. The **Intuition** behind Euler's formula is we can use Taylor series to show that:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Using $x = it$ we get:

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \dots = \cos(t) + i \sin(t)$$

Def. 45. For $z \in \mathbb{C}$, the polar-coordinates are defined as:

$$z = r(\cos(t) + i \sin(t)) = r e^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Rem. 46. To convert from cartesian to polar coordinates:

$$r = \sqrt{a^2 + b^2}$$

$$t = \arctan\left(\frac{b}{a}\right)$$

Note that the arctan function only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so we have to adjust the angle based on the quadrant of the complex number.

Ex. 47. Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1 + t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

Ex. 48. So is exponentiation or taking roots:

$$z^n = (r e^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i \frac{t + 2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Def. 49. The n -th roots of unity are the solutions to $z^n = 1$, which are given by:

$$z_k = e^{i \frac{2\pi k}{n}}, \quad k \in \{0, 1, \dots, n-1\}$$

Thm. 50. The sum of the roots of unity is always zero.

$$\sum_{k=0}^{n-1} e^{i \frac{2\pi k}{n}} = 0$$

Thm. 51. The **fundamental theorem of algebra** states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in complex-conjugate pairs.

1.4 Landau Notation (Asymptotics)

Def. 52. Let f, g be functions.

• $f(x) = O(g(x))$ as $x \rightarrow x_0$ iff $\exists C, \delta > 0$ s.t. $|f(x)| \leq C|g(x)|$ for $|x - x_0| < \delta$.

• $f(x) = o(g(x))$ as $x \rightarrow x_0$ iff $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0$.

• $f(x) \sim g(x)$ as $x \rightarrow x_0$ iff $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1$.

Rem. 53. Useful for Taylor: $f(x) = P_n(x) + O(|x - x_0|^{n+1})$.

1.5 Exam Strategy: Sets and Topology

• **Open:** Every point is an interior point (ball $B_\epsilon(x) \subset A$). Check: Is the boundary excluded?

• **Closed:** Complement is open OR contains all limit points. Check: Is the boundary included?

• **Bounded:** $A \subset B_R(0)$ for some R .

• **Compact:** In $\mathbb{R}^n$, compact $\iff$ closed and bounded.

• **Supremum/Infimum Tips:**

– For $A = \{a_n : n \in \mathbb{N}\}$, check if a_n is monotonic. If $a_n \nearrow L$, then $\sup A = L$, $\inf A = a_1$.

– If a_n oscillates, check even/odd indices separately.

1.6 Exam Strategy: Complex Equations

• **Circles:** $|z - z_0| = R$ is a circle of radius R around z_0 .

• **Lines:** $|z - a| = |z - b|$ is the perpendicular bisector of a and b .

• **Polynomials:** If $P(z) = 0$ and coefficients are real, then $z \in \mathbb{C}$ is a root $\implies \bar{z}$ is also a root.

• **Roots of Unity Sum:** $\sum_{k=0}^{n-1} z_k = 0$ for $n > 1$.

2. Sequences and Series

2.1 Sequences

Def. 54. A **sequence** (Folge) $(a_n)_{n \in \mathbb{N}_0}$ is an infinite list of numbers, mathematically a mapping $f : \mathbb{N}_0 \rightarrow X$. It can be given explicitly or recursively as

$$a_n = f(n) \quad \text{or} \quad a_n = g(a_{n-1}, a_{n-2}, \dots)$$

Def. 55. A sequence (a_n) is called **convergent** with limit $L \in \mathbb{R}$ if:

$$\forall \epsilon > 0 \exists N > 0 \quad \text{s.t.}$$

$$\forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n \rightarrow \infty} a_n = L$. If no such L exists, the sequence is **divergent**.

Rem. 56. A special case of divergent sequences are sequences which diverge to $\pm\infty$. Their limits are called **improper limits**.

Thm. 57. The limit of a sequence is unique aslong as it exists.

Thm. 58. Any subsequence $(a_{n_k})_{k \in \mathbb{N}_0}$ of a convergent sequence converges to the same limit L .

Def. 59. An **accumulation point** $A \in \mathbb{R}$ of a sequence fulfills:

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N$$

$$|a_n - A| < \epsilon$$

Thm. 60. A is an accumulation point $\iff$ there exists a convergent subsequence with limit A .

Thm. 61. Every convergent sequence has exactly one accumulation point (its limit).

Def. 62. A sequence is **upper (lower) bounded** if there exists M such that $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.

Thm. 63. Every convergent sequence is bounded.

Thm. 64. Bolzano-Weierstrass Theorem: Every bounded sequence has an accumulation point.

Thm. 65. Limit arithmetic: For $\lim a_n = K$, $\lim b_n = L$ and $C \in \mathbb{R}$:

$$\lim(a_n \pm b_n) = K \pm L$$

$$\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$$

$$\lim(a_n/b_n) = K/L$$

Rem. 66. Important limits:

- $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$
- $\lim_{n \rightarrow \infty} n^{1/n} = 1$
- $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$
- $\lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0 \quad (a > 1)$
- $\lim_{n \rightarrow \infty} \frac{\log_a n}{n^k} = 0 \quad (a > 1, k > 0)$
- $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$
- $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$

Thm. 67. Growth Hierarchy: For $x \rightarrow \infty$ or $n \rightarrow \infty$:

$$\ln(\ln n) \ll \ln n \ll n^\epsilon \ll n^k \ll a^n \ll n! \ll n^n \quad (\epsilon, k > 0, a > 1)$$

2.2 Convergence Tests for Series

Test	Condition	Conclusion
Null seq.	$\lim a_n \neq 0$	Divergent
Geometric	$\sum q^n$	$ q < 1$ Conv, sum $\frac{1}{1-q}$
Harmonic	$\sum \frac{1}{n^s}$	$s > 1$ Conv, $s \leq 1$ Div
Ratio	$\lim a_{n+1}/a_n = \rho$	$\rho < 1$ Conv, $\rho > 1$ Div
Root	$\lim a_n ^{1/n} = \rho$	$\rho < 1$ Conv, $\rho > 1$ Div
Leibniz	$\sum (-1)^n b_n, b_n \searrow 0$	Convergent
Integral	$a_n = f(n), f \geq 0, \searrow$	$\int_1^\infty f(x)dx$ conv $\iff$
Comp. I	$ a_n \leq b_n$	$\sum b_n$ conv $\implies \sum a_n$ conv
Comp. II	$a_n \geq b_n \geq 0$	$\sum b_n$ div $\implies \sum a_n$ div

Thm. 68. Sandwich-Theorem: If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.

Thm. 69. Useful Inequalities (for Sandwich Theorem & Bounding):

- Bernoulli's Inequality:** $(1+x)^n \geq 1+nx$ (for $x \geq -1$ and $n \in \mathbb{N}$)
- Reverse Triangle Inequality:** $||x| - |y|| \leq |x - y|$
- Trigonometric Bounding:** $\sin(x) \leq x \leq \tan(x)$ (for $x \in (0, \pi/2)$)

Def. 70. A sequence is **monotonically increasing (decreasing)** if $a_n \leq (\geq) a_m$ for $m > n$.

Thm. 71. Monotone Convergence Theorem: Every bounded monotonic sequence converges to its supremum (infimum).

Def. 72. The limit of the supremum is called **limes superior** and denoted as:

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$$

The limit of the infimum is called **limes inferior** and denoted as:

$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k$$

Def. 73. A sequence is a **Cauchy sequence** iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \epsilon$$

Thm. 74. Every Cauchy sequence is bounded.

Thm. 75. A sequence of real numbers converges iff it is a cauchy sequence.

2.3 Series

Def. 76. A **series** is an infinite sum $\sum_{i=0}^\infty a_i$ of the terms of a sequence (a_n) .

Def. 77. A series is said to converge iff the sequence of partialsums converges.

$$\sum_{i=0}^\infty a_i = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_i$$

Rem. 78. Some examples of series are: **geometric series**:

$$\sum_{n=0}^\infty x^n = \frac{1}{1-x}$$

for $|x| < 1$ **harmonic series**:

$$\sum_{n=1}^\infty \frac{1}{n}$$

diverges for $s > 1$

Thm. 79. Necessary condition for convergence: for $\sum_{i=0}^\infty a_i$ to converge we need $\lim_{n \rightarrow \infty} a_n = 0$ i.e. it is a null sequence.

Thm. 80. For **series with non-negative terms** $a_n \geq 0$, the sequence of partial sums is monotonically increasing and **converges iff it is bounded**.

Def. 81. A series is **absolutely convergent** if $\sum |a_n|$ converges. A **conditionally convergent** series converges, but not absolutely.

Thm. 82. Riemann's rearrangement theorem: Let $\sum a_n$ be a conditionally convergent series. Then for any $x \in \mathbb{R}$, there exists a rearrangement of the series that converges to x .

Thm. 83. Comparison test (Vergleichskriterium): If $c_n \leq b_n \leq a_n$ for $n \geq N$: If $\sum a_n$ converges, then $\sum b_n$ converges (**Majorantenkriterium**). If $\sum c_n$ diverges, then $\sum b_n$ diverges (**Minorantenkriterium**).

Thm. 84. Leibniz Criterion: For an alternating series $\sum (-1)^n a_n$ with a monotonically decreasing null sequence (a_n) , the series converges.

Thm. 85. Cauchy Criterion: The series $\sum a_n$ converges iff

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : \left| \sum_{k=n}^m a_k \right| < \epsilon$$

Thm. 86. Root Criterion (Wurzelkriterium): Let $\rho = \limsup \sqrt[n]{|a_n|}$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Thm. 87. Ratio Criterion (Quotientenkriterium): Let $\rho = \lim \left| \frac{a_{n+1}}{a_n} \right|$. If $\rho < 1$, abs conv; if $\rho > 1$, div.

Def. 88. A **power series** (Potenzreihe) centered at a has the form

$$\sum_{k=0}^\infty c_k (x-a)^k$$

Thm. 89. Every power series has a **convergence radius** R , with the **interval of convergence** $(a-R, a+R)$. R is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.

2.4 Exam Strategy: Sequence & Series Convergence

- **Type $\frac{\ln^\alpha n}{n^\beta}$:** Converges for $\beta > 0$ (any α), or $\beta = 0, \alpha < 0$.
- **Type $\sum \frac{1}{n \ln^\alpha n}$ (Bertrand):** Converges $\iff \alpha > 1$.
- **Finding $\limsup / \liminf$:** List accumulation points. If a_n is defined piecewise or with $(-1)^n$, look at sub-sequences.
- **Radius of Convergence R :** Use $R = \lim |\frac{c_n}{c_{n+1}}|$ or $R = \frac{1}{\lim \sqrt[n]{|c_n|}} \in [0, \infty]$.
- **Boundary Check:** Always check $|x - x_0| = R$ separately. Use Leibniz or Comparison.
- **Telescoping Series:** Write $a_n = b_n - b_{n+1}$. Partial sum $S_N = b_1 - b_{N+1}$.

3. Functions

- Def. 90.** A **function** $f : D \rightarrow C$ is a map from a domain D to a codomain C s.t. for each $x \in D$ there is exactly one $y \in C$ s.t. $f(x) = y$ (well defined).
- Def. 91.** A function f is called **bounded** if there exists $M \in \mathbb{R}$ such that $|f(x)| \leq (\geq) M$ for all $x \in D$.
- Def. 92.** A function f is called **injective** iff
- $$\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$$
- surjective** iff
- $$\forall y \in B \exists x \in A : f(x) = y$$
- and **bijective** iff f is injective and surjective.
- Def. 93.** The **restriction** of f under $D' \subset D$ is:
- $$f \upharpoonright_{D'} : D' \rightarrow C, x \mapsto f(x) \quad \forall x \in D$$
- Def. 94.** Let $f : D \rightarrow C$ and $g : C \rightarrow B$ be functions then
- $$g \circ f : D \rightarrow B \quad \text{with}$$
- $$g \circ f(x) = g(f(x)) \quad \forall x \in D$$
- is called the **composition** of f and g .
- Def. 95.** The **identity function** $id_D : D \rightarrow D := id_D(x) = x \quad \forall x \in D$.

- Def. 96.** If a function $f : D \rightarrow C$ is bijective then there exists a unique function $f^{-1} : C \rightarrow D$ such that
- $$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$
- and
- $$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$
- f^{-1} is called the **inverse function** of f .
- Def. 97.** A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called **even** iff
- $$\forall x \in \mathbb{R} : f(-x) = f(x)$$
- and **odd** iff
- $$\forall x \in \mathbb{R} : f(-x) = -f(x)$$
- Def. 98.** A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called **(strictly) monotonically increasing** iff
- $$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$$
- and **(strictly) monotonically decreasing** iff
- $$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$$
- Thm. 99.** Every strictly monotonic function is injective.
- 3.1 Limits in the context of functions**
- Def. 100.** Let $f : \mathbb{D} \rightarrow \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \rightarrow x_0} f(x) = L$ iff
- $$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t.}$$
- $$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$
- The limit L is unique.
- Def. 101.** A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a **right-sided limit** $\lim_{x \searrow x_0} f(x) = \lim_{x \rightarrow x_0^+} f(x) = L$ iff
- $$\forall \varepsilon > 0 \exists \delta > 0 \quad \text{s.t.}$$
- $$\forall x \in \mathbb{D}(x_0 < x < x_0 + \delta) \implies |f(x) - L| < \varepsilon$$
- Def. 102.** A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has a **limit for x approaching infinity** $\lim_{x \rightarrow \infty} f(x) = L$ iff
- $$\forall \varepsilon > 0 \exists M > 0 \quad \text{s.t.}$$
- $$\forall x \in \mathbb{D}(x > M) \implies |f(x) - L| < \varepsilon$$

- Def. 103.** A function $f : \mathbb{D} \rightarrow \mathbb{R}$ has in x_0 a **improper limit approaching infinity** $\lim_{x \rightarrow x_0} f(x) = \infty$ iff
- $$\forall N > 0 \exists \delta > 0 \quad \text{s.t.}$$
- $$\text{for all } x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$
- Thm. 104.** Let $\lim_{x \rightarrow x_0} f(x) = K (\neq \pm\infty)$ and $\lim_{x \rightarrow x_0} g(x) = L (\neq \pm\infty)$
- $$\lim_{x \rightarrow x_0} (f(x) + g(x)) = K + L$$
- $$\lim_{x \rightarrow x_0} (f(x) - g(x)) = K - L$$
- $$\lim_{x \rightarrow x_0} (f(x)g(x)) = KL$$
- $$\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L$$
- $$\lim_{x \rightarrow x_0} (Af(x)) = AK$$
- $$\lim_{x \rightarrow x_0} (f(x)/g(x)) = K/L \text{ if } L \neq 0$$
- 3.2 Continuity of functions**
- Def. 105.** A function $f : D \rightarrow \mathbb{R}$ is called **continuous** at x_0 if
- $$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that}$$
- $$\forall x \in I(|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon)$$
- The function is called continuous if it is continuous at every point in its domain.
- Rem. 106.** The **intuition** behind this definition is that for any given distance $\varepsilon > 0$ around $f(x_0)$, we can find a $\delta > 0$ such that for all $x \in D$ that are at least δ close to x_0 , we have $|f(x) - f(x_0)| < \varepsilon$. So for any bound on the output (ε) there exists a bound on the input (δ) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(f(x_0) - \varepsilon, f(x_0) + \varepsilon)$.
- Def. 107.** Let $f : D \rightarrow \mathbb{R}$ be a function. Then f is called **left(right)-continuous** at x_0 if
- $$\lim_{x \rightarrow x_0^{-(+)}} f(x) = f(x_0)$$
- Thm. 108.** Let $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ be a function and $x_0 \in D$. f is continuous at x_0 iff for every sequence $(x_n)_{n \in \mathbb{N}} \subset D$ with $\lim_{n \rightarrow \infty} x_n = x_0$ it holds that $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$.

- Rem. 109.** Continuity can also be viewed in terms of intervals, where a function f is continuous iff for every open interval $I \subset \mathbb{R}$ it holds that $f^{-1}(I)$ is open.
- Lem. 110.** Let f, g be continuous functions then
- $$f + g, \quad f - g, \quad f \cdot g, \quad c \cdot f, \quad \frac{f}{g}, \quad f \circ g$$
- are continuous
- Thm. 111.** Let I be an interval and let $f : I \rightarrow J$ be a continuous and bijective function. Then J is an interval too and the inverse function $f^{-1} : J \rightarrow I$ is continuous.
- Thm. 112.** Let $f : D \rightarrow \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.
- Lem. 113.** Let $[a, b]$ be a compact interval and let $(x_n)_{n \in \mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that
- $$\lim_{k \rightarrow \infty} x_{n_k} = x_0 \quad \text{for some } x_0 \in [a, b].$$
- Def. 114.** Let $f : D \subset \mathbb{R} \rightarrow \mathbb{R}$ be a function. If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then f has a local maximum at x_0 . If for $x_0 \in D$ it holds that $f(x_0) \leq f(x) \forall x \in D$, then f has a local minimum at x_0 .
- Thm. 115.** Let $f : I \subset \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and I be a compact interval. Then f is bounded and attains its maximum and minimum value on I . Hence
- $$\exists x_{min}, x_{max} \in I \text{ such that}$$
- $$f(x_{min}) \leq f(x) \leq f(x_{max}) \quad \forall x \in I$$
- 3.3 Trigonometric Identities**
- | Category | Formula |
|--------------|---|
| Pythagoras | $\sin^2 x + \cos^2 x = 1$
$1 + \tan^2 x = \frac{1}{\cos^2 x}$ |
| Addition | $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ |
| Double Angle | $\sin(2x) = 2 \sin x \cos x$
$\cos(2x) = \cos^2 x - \sin^2 x$
$= 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$ |
| Half Angle | $\sin^2 x = \frac{1 - \cos(2x)}{2}$
$\cos^2 x = \frac{1 + \cos(2x)}{2}$ |
| Subst. trick | $\tan(x/2) = t \implies$
$\sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}$
$dx = \frac{2dt}{1+t^2}$ |
| Prod-to-Sum | $\sin x \sin y = \frac{1}{2}(\cos(x-y) - \cos(x+y))$
$\cos x \cos y = \frac{1}{2}(\cos(x-y) + \cos(x+y))$
$\sin x \cos y = \frac{1}{2}(\sin(x+y) + \sin(x-y))$ |

3.4 Hyperbolic Functions

- $\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}$
- $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- $\cosh^2 x - \sinh^2 x = 1$
- $\sinh(2x) = 2 \sinh x \cosh x, \quad \cosh(2x) = \cosh^2 x + \sinh^2 x$
- $(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x, \quad (\tanh x)' = \frac{1}{\cosh^2 x}$
- $\operatorname{arsinh}(x) = \ln(x + \sqrt{x^2 + 1}), \quad \operatorname{arcosh}(x) = \ln(x + \sqrt{x^2 - 1})$

4. Differential calculus

Def. 116. The **average rate of change** of a function f in the domain between a and $a + h$:=

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}$$

and also called the **difference quotient**.

Def. 117. The **derivative** f' of a function f at the coordinate a :=

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} &= \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} \\ &= \frac{df}{dx} \Big|_{x=a} = f'(a) \end{aligned}$$

Rem. 118. The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of f at the point $(a, f(a))$.

Def. 119. If for a function f at the coordinate a the derivative $f'(a)$ exists then f is called **differentiable** at a . If f is differentiable for every $a \in \mathbb{D}$ then f is called differentiable.

Lem. 120. If f is differentiable at x_0 then f is continuous at x_0 .

Def. 121. If the limit

$$\lim_{h > 0 \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists then we call it the **right-sided derivative** (or the left-sided derivative analogue)

Def. 122. For a function f we can iteratively define higher derivatives for $n \in \mathbb{N}_0$

$$f^{(1)} = f', f^{(2)} = f'', \dots, f^{(n+1)} = (f^{(n)})'$$

if $f^{(n)}$ exists we call it **n -times differentiable**.

Def. 123. if $f^{(n)}$ is continuous we call f **n -times continuous differentiable**. The set of all n -times continuous differentiable functions on D is denoted as $C^n(D)$

Def. 124. If for a function f and some $x_0 \in \mathbb{D}$ and $\delta > 0$ we have

$$f(x_0) \geq (<) f(x) \quad \forall x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta)$$

then x_0 is called a **local maximum (minimum)**.

4.1 Derivation rules

Thm. 125. Common Derivatives:

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^p	px^{p-1}	$\ln x $	$1/x$
e^x	e^x	a^x	$a^x \ln a$
$\sin x$	$\cos x$	$\cos x$	$-\sin x$
$\tan x$	$1 + \tan^2 x$ $= \cos^{-2} x$	$\cot x$	$-(1 + \cot^2 x)$ $= -\sin^{-2} x$
$\sinh x$	$\cosh x$	$\cosh x$	$\sinh x$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$\arccos x$	$\frac{-1}{\sqrt{1-x^2}}$
$\arctan x$	$\frac{1}{1+x^2}$	$\log_a x$	$\frac{1}{x \ln a}$
$\operatorname{arsinh} x$	$\frac{1}{\sqrt{1+x^2}}$	$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x^2-1}}$

Thm. 126. Sum Rule: For the n -th derivative of a sum:

$$(f + g)^{(n)} = f^{(n)} + g^{(n)}$$

Thm. 127. General Leibniz Rule (Product): The n -th derivative of a product is given by:

$$(f \cdot g)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}$$

Thm. 128. Chain Rule: For the composition of functions:

$$\begin{aligned} (f \circ g)'(x) \\ = f'(g(x)) \cdot g'(x) \end{aligned}$$

Thm. 129. Quotient Rule: For $g(x) \neq 0$:

$$\begin{aligned} \left(\frac{f}{g} \right)' \\ = \frac{f'g - fg'}{g^2} \end{aligned}$$

Thm. 130. Inverse Functions: If f is differentiable and injective:

$$\begin{aligned} (f^{-1})'(x) \\ = \frac{1}{f'(f^{-1}(x))} \end{aligned}$$

4.2 Bernoulli-L'Hopital

Thm. 131. Let $I \subset \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$, and let x_0 be a local extremum of f . Then one of the following holds:

- x_0 is an endpoint of I
- f is not differentiable at x_0
- f is differentiable at x_0 and $f'(x_0) = 0$

Thm. 132. Rolle's Theorem: Let $a < b$ with $f(a) = f(b)$, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$f'(\xi) = 0$$

Thm. 133. Mean-Value-Theorem: Let $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

Thm. 134. Cauchy's Mean-Value-Theorem: Let $a < b$, and let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then $\exists \xi \in (a, b)$ with

$$\begin{aligned} \frac{f(b) - f(a)}{g(b) - g(a)} \\ = \frac{f'(\xi)}{g'(\xi)} \end{aligned}$$

Thm. 135. Bernoulli-L'Hopital's Rule: Let $f, g : I \rightarrow \mathbb{R}$ be differentiable functions on (a, b) and let $x_0 \in (a, b)$. If $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$. If

$$\lim_{x \rightarrow x_0} |f(x)| = \lim_{x \rightarrow x_0} |g(x)| \in \{0, \infty\}$$

$$\text{and } \exists L = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$$

then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L$$

4.3 Convex and Concave graphs

Def. 136. A function $f : I \rightarrow \mathbb{R}$ is called **convex** iff $\forall a, b \in I \forall t \in (0, 1)$

$$f((1-t)a + bt) \leq (1-t)f(a) + tf(b)$$

Def. 137. A function $f : I \rightarrow \mathbb{R}$ is called **concave** iff $\forall a, b \in I \forall t \in (0, 1)$

$$f((1-t)a + bt) \geq (1-t)f(a) + tf(b)$$

Thm. 138. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then $\forall x$

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

Thm. 139. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then

$$f \text{ is convex} \iff f' \text{ is increasing}$$

Thm. 140. Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable. Then

$$f \text{ is convex} \iff f'' \geq 0$$

4.4 Taylor Series

Lem. 141. We can locally approximate the graph of f at x_0 with the tangent line $t(x)$:

$$f(x) \approx t(x)$$

$$= f(x_0) + f'(x_0)(x - x_0)$$

Lem. 142. To even more accurately approximate the function f we can add further terms to the tangent line expansion:

$$f(x) \approx t(x)$$

$$= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Thm. 143. Taylor's Theorem: Let $f : (a, b) \rightarrow \mathbb{R}$ be $n + 1$ times differentiable. Then

$$f(x) = P_n(x) + R(x)$$

with $P_n(x)$ the Taylor polynomial of degree n and $R(x)$ the remainder term:

$$\begin{aligned} P_n(x) &= \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \\ R(x) &= \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \quad (\xi \text{ between } x_0, x) \end{aligned}$$

Def. 144. The **Taylor series expansion** of an infinitely differentiable function $f : (a, b) \rightarrow \mathbb{R}$ at x_0 is given by:

$$\begin{aligned} f(x) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \end{aligned}$$

Thm. 145. Standard Taylor Series (at $x = 0$):

Func.	Series	First Terms	Rad
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$1 + x + \frac{x^2}{2} + \dots$	∞
$\sin x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	$x - \frac{x^3}{6} + \dots$	∞
$\cos x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$1 - \frac{x^2}{2} + \dots$	∞
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$1 + x + x^2 + \dots$	1
$\ln(1+x)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$	$x - \frac{x^2}{2} + \dots$	1
$\arctan x$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	$x - \frac{x^3}{3} + \dots$	1
$(1+x)^\alpha$	$\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$	$1 + \alpha x + \dots$	1
$\sinh x$	$\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$	$x + \frac{x^3}{6} + \dots$	∞
$\cosh x$	$\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$	$1 + \frac{x^2}{2} + \dots$	∞

Imp. 146. Radii of Convergence for additional Taylor Series:

- Logarithm:** $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$ (for $x \in (-1, 1]$)
- Geometric:** $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ (for $|x| < 1$)
- Binomial:** $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ (for $|x| < 1$)

Thm. 147. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$ be two power series with radius of convergence R . Then:

$$\begin{aligned} f(x)g(x) &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n \end{aligned}$$

Thm. 148. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R . Then:

$$\begin{aligned} f'(x) &= \sum_{n=1}^{\infty} n a_n x^{n-1} \end{aligned}$$

Thm. 149. Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R . Then:

$$\begin{aligned} \int f(x) dx &= \sum_{n=0}^{\infty} \int a_n x^n dx \end{aligned}$$

4.5 Exam Strategy: Piecewise Functions

To check continuity and differentiability at x_0 :

- Continuity:** Check if $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = f(x_0)$.
- Differentiability:** If continuous, check if $\lim_{h \searrow 0} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h \nearrow 0} \frac{f(x_0+h) - f(x_0)}{h}$.
- Shortcut:** If $f'(x)$ exists for $x \neq x_0$ and $\lim_{x \rightarrow x_0} f'(x) = L$ exists, then $f'(x_0) = L$.

4.6 Exam Strategy: Limits - Taylor vs. L'Hôpital

- Use Taylor when:** The expression has complicated products, powers, or nested functions (e.g., $e^{\sin x} - 1$).
- Use L'Hôpital when:** The expression is a simple fraction $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and derivatives are easy.
- Important Continuous Limits:**

$$\begin{aligned} - \lim_{x \rightarrow 0} \frac{\sin(x)}{x} &= 1 \\ - \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} &= \frac{1}{2} \end{aligned}$$

$$- \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} \right)^x = e^a$$

• **L'Hôpital Algebraic Conversions:**

- $0 \cdot \infty \rightarrow$ rewrite as $\frac{1}{\infty^0}$ or $\frac{1}{0^\infty}$ (i.e. $\frac{f}{1/g}$ or $\frac{g}{1/f}$).
- $\infty - \infty \rightarrow$ combine into a single fraction.
- $1^\infty, 0^0, \infty^0 \rightarrow$ use $L = \lim f(x)^{g(x)} \implies \ln(L) = \lim [g(x) \ln(f(x))]$.

5. Integration

5.1 Indefinite Integrals and Techniques

Def. 150. Functions $F : I \rightarrow \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in (a, b)$ are called **antiderivatives** of f .

Def. 151. The family of all anti-derivatives of a function $f : I \rightarrow \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval is called the **indefinite integral** and $:=$

$$\begin{aligned} \int f(x) dx &= \{F : I \rightarrow \mathbb{R} : F'(x) = f(x)\} = F(x) + C \end{aligned}$$

Thm. 152. Let $f, g : I \rightarrow \mathbb{R}$ be integrable functions and let $c \in \mathbb{R}$. Then:

$$\begin{aligned} \int (f(x) \pm g(x)) dx &= \int f(x) dx \pm \int g(x) dx \\ \int cf(x) dx &= c \int f(x) dx \end{aligned}$$

Thm. 153. Common Antiderivatives:

$f(x)$	$\int f(x) dx$	$f(x)$	$\int f(x) dx$
x^p	$\frac{x^{p+1}}{p+1}$ ($p \neq -1$)	$\frac{1}{x}$	$\ln x $
e^x	e^x	$\frac{1}{a^x}$	$\frac{a^{-x}}{\ln a}$
$\sin x$	$-\cos x$	$\cos x$	$\sin x$
$\tan x$	$-\ln \cos x $	$\frac{1}{\cos^2 x}$	$\tan x$
$\sinh x$	$\cosh x$	$\cosh x$	$\sinh x$
$\frac{1}{1+x^2}$	$\arctan x$	$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x$
$\frac{1}{\sqrt{x^2+1}}$	$\operatorname{arsinh} x$	$\frac{1}{\sqrt{x^2-1}}$	$\operatorname{arcosh} x$
$\ln x$	$x \ln x - x$	$\frac{1}{x^2-1}$	$\frac{1}{2} \ln \left \frac{x-1}{x+1} \right $

Rem. 154. Integration Tricks:

- Partial Fraction Decomposition:** For $\int \frac{P(x)}{Q(x)} dx$. If $\deg P \geq \deg Q$, use polynomial division first.
 - Simple real root: $(ax+b) \rightarrow \frac{A}{ax+b}$
 - Repeated real root: $(ax+b)^k \rightarrow \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_k}{(ax+b)^k}$
 - Irreducible quadratic: $(ax^2+bx+c) \rightarrow \frac{Ax+B}{ax^2+bx+c}$
- Weierstrass Substitution:** For integrals of $R(\sin x, \cos x)$, let $t = \tan(x/2)$.

- Square Completion:** For $\sqrt{ax^2+bx+c}$, complete the square to get $\sqrt{u^2 \pm k^2}$.

• **Substitution Patterns:**

$$\begin{aligned} - \sqrt{a^2 - x^2} &\rightarrow x = a \sin \theta \text{ or } x = a \cos \theta \\ - \sqrt{a^2 + x^2} &\rightarrow x = a \tan \theta \text{ or } x = a \sinh u \\ - \sqrt{x^2 - a^2} &\rightarrow x = a \sec \theta \text{ or } x = a \cosh u \end{aligned}$$

- Symmetry:** $\int_{-a}^a f(x) dx = 0$ if f is odd, and $2 \int_0^a f(x) dx$ if f is even.

Thm. 155. Integration by Parts: For differentiable functions $f(x)$ and $g(x)$ we have:

$$\begin{aligned} \int f'(x)g(x) dx &= f(x)g(x) - \int f(x)g'(x) dx \end{aligned}$$

Thm. 156. Substitution Rule: For functions $f(x)$ and $g(x)$ we have:

$$\begin{aligned} \int f(g(x))g'(x) dx &= \int f(u) du = F(g(x)) + C \end{aligned}$$

where F is an antiderivative of f and $u = g(x)$.

5.2 The Definite Integral (Riemann Integral)

Def. 157. Let U and L denote the upper and lower sums of a bounded function f . Then f is called **Riemann-integrable** iff $\inf U = \sup L$. The value of the definite integral $\int_a^b f(x) dx$ is then this common value.

Thm. 158. Integrability Conditions: Any continuous or monotone function on $[a, b]$ is Riemann-integrable.

Thm. 159. Properties of Riemann Integral: Let f, g be integrable and $c \in \mathbb{R}$:

$$\begin{aligned}\int_a^b (f(x) \pm g(x)) dx &= \int_a^b f(x) dx \pm \int_a^b g(x) dx \\ \int_a^b cf(x) dx &= c \int_a^b f(x) dx \\ f(x) \leq g(x) &\implies \int_a^b f(x) dx \leq \int_a^b g(x) dx \\ \left| \int_a^b f(x) dx \right| &\leq \int_a^b |f(x)| dx \\ \int_a^b f(x) dx &= - \int_b^a f(x) dx \\ \int_a^c f(x) dx &= \int_a^b f(x) dx + \int_b^c f(x) dx\end{aligned}$$

Thm. 160. Mean Value Theorem for Integrals: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then there exists a $c \in [a, b]$ such that:

$$\int_a^b f(x) dx = (b - a)f(c)$$

Thm. 161. Fundamental Theorem of Integral Calculus: Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then for all $C \in \mathbb{R}$, $F(x)$ is an antiderivative of f where:

$$F(x) = \int_a^x f(y) dy + C$$

5.3 Sequences of Functions and Integrals

Def. 162. Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of functions. **Pointwise Convergence**: It converges pointwise to f if for every x , the sequence of real numbers $(f_n(x))$ converges to $f(x)$.

Def. 163. Uniform Convergence: It converges uniformly to f if for every $\epsilon > 0$, there exists an N such that for all $n \geq N$ and for all x simultaneously, $|f_n(x) - f(x)| < \epsilon$.

Thm. 164. Swapping Limit and Integral: If a sequence of integrable functions (f_n) converges **uniformly** to f on $[a, b]$, then the limit function f is also integrable, and:

$$\begin{aligned}\int_a^b f(x) dx \\ = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx\end{aligned}$$

5.4 Definite Integration Techniques

Lem. 165. If $F : [a, b] \rightarrow \mathbb{R}$ is continuously differentiable, then:

$$\begin{aligned}F(x) &= F(a) + \int_a^x F'(t) dt \\ \int_a^b F'(t) dt &= F(b) - F(a) = \left[F(t) \right]_a^b\end{aligned}$$

Lem. 166. Substitution for Definite Integrals: Let f be continuous and g continuously differentiable.

$$\begin{aligned}\int_a^b f(g(x))g'(x) dx \\ = \int_{g(a)}^{g(b)} f(u) du\end{aligned}$$

Version 2: If $f'(x) \neq 0$ and f^{-1} is the inverse of f :

$$\begin{aligned}\int_a^b g(f(x)) dx \\ = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y) dy\end{aligned}$$

Lem. 167. Integration by Parts for Definite Integrals: Let f, g be continuously differentiable.

$$\begin{aligned}\int_a^b f'(x)g(x) dx \\ = \left[f(x)g(x) \right]_a^b - \int_a^b f(x)g'(x) dx\end{aligned}$$

5.5 Exam Strategy: Leibniz's Rule

$$\begin{aligned}\frac{d}{dx} \int_{a(x)}^{b(x)} f(t, x) dt &= f(b(x), x)b'(x) \\ - f(a(x), x)a'(x) &+ \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x} dt\end{aligned}$$

5.6 Geometric Applications

- **Area between curves**: $A = \int_a^b |f(x) - g(x)| dx$.
- **Rotational Volume (around x-axis)**: $V = \pi \int_a^b (f(x))^2 dx$.
- **Rotational Volume (around y-axis)**: $V = 2\pi \int_a^b xf(x) dx$.
- **Arc Length**: $L = \int_a^b \sqrt{1 + (f'(x))^2} dx$.

5.7 Convergence of Improper Integrals

- **p-test (to infinity)**: $\int_1^\infty \frac{1}{x^p} dx$ converges if $p > 1$ (diverges if $p \leq 1$).
- **p-test (to zero)**: $\int_0^1 \frac{1}{x^p} dx$ converges if $p < 1$ (diverges if $p \geq 1$).
- **Comparison Test**: If $0 \leq f(x) \leq g(x)$, then $\int g$ conv $\implies \int f$ conv.

6. Differential Equations

6.1 Modelling and Definitions

Rem. 168. Modelling Process: To derive an ODE from a physical scenario:

1. Identify the quantity $u(t)$ and express its change over a small interval Δt : $u(t + \Delta t) = u(t) + (\text{rate}) \cdot \Delta t$
2. Rearrange and take the limit $\Delta t \rightarrow 0$: $\frac{u(t+\Delta t) - u(t)}{\Delta t} = \text{rate} \implies u'(t) = \text{rate}$

Rem. 169. Typical ODE Models:

1. Bounded Growth (Logistic): With capacity L :
$$u'(t) = ku(t) \left(1 - \frac{u(t)}{L} \right)$$
2. Bimolecular Reaction ($A + B \rightarrow C$): With concentrations a, b :
$$u'(t) = k(a - u(t))(b - u(t))$$

Def. 170. An ordinary differential equation (ODE) contains derivatives of a function of one independent variable. It is:

- **of n-th order**: if the n -th derivative is the highest occurring.
- **linear**: if the unknown function $u(t)$ and its derivatives appear only as a linear combination.
- **homogeneous**: if it only contains terms with $u(t)$ and its derivatives (otherwise inhomogeneous).

Thm. 171. Picard-Lindelöf (Existence and Uniqueness): If $f(x, y)$ is continuous and Lipschitz continuous with respect to y , then the IVP $y' = f(x, y)$ with $y(x_0) = y_0$ has a unique solution in some interval around x_0 .

6.2 First-Order ODEs

Thm. 172. Separation of Variables: For $y' = f(x)g(y) \implies \int \frac{dy}{g(y)} = \int f(x) dx + C$.

Thm. 173. Linear First-Order ODE: $y' + p(x)y = q(x)$. **Integrating Factor Method**: 1. Calculate $F(x) = \exp\left(\int p(x) dx\right)$. 2. Multiply ODE by $F(x)$: $(yF(x))' = q(x)F(x)$. 3. Integrate: $y(x) = \frac{1}{F(x)} \left(\int q(x)F(x) dx + C \right)$.

Thm. 174. Bernoulli ODE: $y' + p(x)y = q(x)y^n$ ($n \neq 0, 1$). **Trick**: Substitute $u = y^{1-n} \implies u' = (1-n)y^{-n}y'$. This transforms the ODE into the linear form: $\frac{u'}{1-n} + p(x)u = q(x)$.

Thm. 175. Homogeneous substitution: $y' = f(y/x)$. Substitute $u = y/x \implies y = ux, y' = u'x + u$.

6.3 Linear ODEs of n-th Order

Thm. 176. Superposition Principle: If y_1 and y_2 solve a linear homogeneous ODE, then for any $C_1, C_2 \in \mathbb{R}$:

y(x) = C1y1(x) + C2y2(x)

Thm. 177. An n-th order linear homogeneous ODE has n linearly independent solutions $y_1, \dots, y_n$. Its general solution is:

y(x) = \sum_{i=1}^n Ci yi(x) with Ci \in \mathbb{R}

Lem. 178. For a homogeneous ODE with constant coefficients $a_n y^{(n)} + \dots + a_0 y = 0$, the Ansatz $y(x) = e^{\lambda x}$ yields the characteristic equation:

an \lambda^n + an-1 \lambda^{n-1} + \dots + a1 \lambda + a0 = 0

with characteristic polynomial $p(\lambda)$.

Thm. 179. General Solution from Characteristic Roots: The roots of $p(\lambda)$ determine the fundamental solutions:

- Distinct real root $\lambda \in \mathbb{R}$

y(x) = e^{\lambda x}

- Multiple real root $\lambda \in \mathbb{R}$ with multiplicity m

e^{\lambda x}, x e^{\lambda x}, \dots, x^{m-1} e^{\lambda x}

- Complex pair $\lambda = \alpha \pm i\beta \in \mathbb{C}$ (multiplicity m)

e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x), \dots,

x^{m-1} e^{\alpha x} \cos(\beta x), x^{m-1} e^{\alpha x} \sin(\beta x)

6.4 Inhomogeneous Linear ODEs

Lem. 180. The general solution of $a_n y^{(n)} + \dots + a_0 y = s(x)$ is $y(x) = y_h(x) + y_p(x)$, where y_h is the general homogeneous solution and y_p is a particular solution.

6.4.1 Method of Undetermined Coefficients (Ansatz Table)

For $ay'' + by' + cy = s(x)$, if no resonance:

Source $s(x)$	Ansatz $y_p(x)$
$P_n(x)$ (Polyn.)	$Q_n(x) = A_n x^n + \dots + A_0$
$A e^{kx}$	$B e^{kx}$
$A \sin(\omega x) + B \cos(\omega x)$	$C \sin(\omega x) + D \cos(\omega x)$
$e^{\alpha x} P_n(x)$	$e^{\alpha x} Q_n(x)$
$e^{\alpha x} \sin(\beta x)$	$e^{\alpha x} (C \cos(\beta x) + D \sin(\beta x))$

Resonance Rule: If the standard Ansatz $y_p(x)$ contains a solution to the homogeneous equation, multiply the entire Ansatz by x (or x^2 for double roots).

Imp. 181. Resonance: If the term e^{kx} or $e^{i\omega x}$ in $s(x)$ matches an m -fold root of the characteristic polynomial $p(\lambda)$, multiply the standard Ansatz $y_p(x)$ by x^m .

6.4.2 Variation of Constants

Thm. 182. Variation of Constants (First-Order Linear ODE): For $y' + p(x)y = q(x)$, the general solution is:

y(x) = e^{-P(x)} \left(\int q(x) e^{P(x)} dx + C \right)

where $P(x) = \int p(x) dx$ is an antiderivative of $p(x)$.

6.5 Initial Value Problems and Boundary-Value Problems

Def. 183. An Initial Value Problem (IVP) specifies conditions at a single point x_0 (e.g. $y(x_0) = y_0, y'(x_0) = y_1$). A Boundary Value Problem (BVP) specifies conditions at multiple points (e.g. $y(a) = y_a, y(b) = y_b$).

6.6 Exam Strategy: ODE Solver Workflow

- Can I separate? $y' = f(x)g(y)$.
- Is it linear 1st order? $y' + p(x)y = q(x) \rightarrow$ Integrating factor $e^{\int p}$.
- Is it 2nd order linear with constant coeffs? $ay'' + by' + cy = s(x)$.
 - Solve char. eq. $a\lambda^2 + b\lambda + c = 0$.
 - Particular solution: Match $s(x)$ with Ansatz table (check for resonance!).

6.7 Exam Strategy: Constrained Extrema

To find extrema of $f(x, y)$ s.t. $g(x, y) = 0$:

- Set up system $\nabla f = \lambda \nabla g$ and $g = 0$.

- Solve for x, y, λ .
- Don't forget the boundary! If the region is a disk $x^2 + y^2 \leq 1$, check interior $\nabla f = 0$ AND boundary $x^2 + y^2 = 1$.